An Assessment of Shells

Resolving the Unit-Width vs Dyadic Convention
in the Angular Cancellation Programme

A Definitive Reference Settling the Shell Decomposition Question
with Evidence from Harmonic Analysis, PDE Theory,
Formal Verification, and AI Governance

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Purpose. The shell decomposition convention — unit-width versus dyadic — was flagged as a concern by every AI system that audited the Angular Cancellation manuscript. Three systems, multiple sessions, the same objection every time. This document settles the question permanently. It is simultaneously a mathematical reference, a corrections record, a formal verification explainer, an AI governance finding, and a closure statement. After this document, the shell convention question is closed.

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1 The Problem Statement

During the adversarial AI audit of the Angular Cancellation manuscript, the shell decomposition convention was raised as a concern independently by every participating system. The objections took two forms:

1. **Paper–Coq mismatch.** The manuscript defines unit-width shells $S_j = \{k \in \mathbb{Z}^3 : j \leq |k| < j + 1\}$, while the Coq formalisation contains a definition `in_shell` using dyadic bands on $|k|^2$. Systems flagged this as a "specification mismatch," "inconsistency," or "mathematical error."

2. **Appendix convention drift.** Several appendix computations (C.7, D.4.10, G.1, G.7) used dyadic series constants or dyadic shell definitions in a paper whose main argument uses unit-width shells. Systems flagged these as notation errors or proof gaps.

The first objection is wrong. The second is a genuine notation inconsistency that has been corrected. This document explains why the first is wrong, documents the corrections for the second, and settles both issues with references to the mathematical and formal verification literature.

The shell convention was raised in:

- The initial ACL audit (all three systems)
- The Section 6 review (one system)
- The Remark 9.7 review (one system)
- The Appendix C review (two systems)
- The Appendix G review (one system)
- The Coq formalisation review (all three systems)

No other single issue was raised as frequently.

2 The Two Conventions Defined

2.1 Unit-Width Shells

$$S_j = \{k \in \mathbb{Z}^3 : j \leq |k| < j + 1\}, \qquad k_j = j, \qquad \Delta k = 1. \tag{1}$$

The shell cardinality satisfies

$$|S_j| = 4\pi j^2 + O(j) + O(j^\alpha),$$

where α is the error exponent in the Gauss sphere problem. The leading term $4\pi j^2$ is the surface area of a sphere of radius j — a unit-width shell at radius j sweeps out approximately this surface area on $\mathbb{Z}^3$.

The error exponent has been progressively improved:

Author(s)	Year	Exponent α
Trivial (surface area)	—	2
Poisson summation	—	3/2
Vinogradov; Chen	1963	$< 3/2$
Chamizo–Iwaniec	1995	$29/22 \approx 1.318$
Heath-Brown	1999	$\mathbf{21/16 = 1.3125}$
Conjectured	—	$1/2 + \varepsilon$

The fine arithmetic structure within each shell is governed by the representation function $r_3(n) = \#\{(a, b, c) \in \mathbb{Z}^3 : a^2 + b^2 + c^2 = n\}$. By the Gauss–Legendre theorem, $r_3(n) = 0$ if and only if n has the form $4^a(8b + 7)$. The individual $r_3(n)$ values exhibit wild arithmetic fluctuations connected to class numbers of imaginary quadratic fields and weight-3/2 modular forms. Within a unit-width shell, these fluctuations remain visible as significant relative perturbations — the error-to-main-term ratio decays only as $O(j^{\alpha-2}) \approx O(j^{-0.69})$.

2.2 Dyadic (Littlewood–Paley) Shells

$$S_j = \{k \in \mathbb{Z}^3 : 2^{j-1} \leq |k| < 2^j\}, \qquad k_j = 2^j, \qquad \Delta k = 2^{j-1}. \tag{2}$$

The shell cardinality scales as $|S_j| \sim (7\pi/6) \cdot 2^{3j}$ (volume scaling, not 2^{2j}). A dyadic shell of width 2^{j-1} at radius 2^j is a thick three-dimensional region whose lattice content scales with its volume.

Because a dyadic shell spans $\sim (3/4) \cdot 4^j$ consecutive values of $|k|^2$, the representation-theoretic fluctuations of $r_3(n)$ average out by a law-of-large-numbers effect. The error-to-main-term ratio decays as $O(2^{-27j/16})$ — *exponentially* small. Arithmetic structure is completely invisible at dyadic resolution.

2.3 What Each Convention Makes Visible

Feature	Unit-Width	Dyadic
Lattice arithmetic ($r_3(n)$)	Visible	Averaged out
Equidistribution on thin shells	Accessible	Inaccessible
Transverse-collinear decomposition	Natural	Unnatural
Per-mode geometric arguments	Preserved	Lost
LP square function equivalence	One-sided only	Full two-sided
Bernstein inequalities	Degenerate	Uniform
Bony paraproduct	Not available	Standard

3 Why This Paper Uses Unit-Width Shells

The Angular Cancellation Lemma depends on three geometric arguments that require lattice-resolution information:

1. **Transverse-collinear decomposition (Definition 4.8, Lemma 4.22).** The resonant set $\mathcal{R}_j(k)$ is partitioned into modes with $|p_\perp| \geq c_0|k|$ (transverse) and $|p_\perp| < c_0|k|$ (collinear). The counting bound $\#\mathcal{C}^{\text{coll}} \leq Cc_0k_j$ depends on the number of lattice points in a thin tube of radius c_0k_j and length $O(k_j)$. At unit-width resolution, this is a sharp lattice geometry argument. At dyadic resolution, the tube spans $O(2^{j-1})$ unit shells and the counting bound loses its geometric content.
2. **Equidistribution on thin spherical shells (Appendix G, Theorem .26).** The lattice equidistribution theorem (Duke–Linnik–Schulze-Pillot) applies to lattice points on thin spherical shells. The theorem requires the shell thickness to be $O(1)$ relative to the lattice spacing. A dyadic shell of thickness 2^{j-1} does not satisfy this condition for $j \geq 2$.
3. **Shell cardinality in the Cauchy–Schwarz step (Theorem 4.27, Step 3).** The ACL's $k_j^{1/2}$ saving comes from Cauchy–Schwarz over the transverse set, which introduces a factor $(\#\mathcal{C}^{\text{trans}})^{1/2} \sim k_j$. The shell cardinality $|S_j| \sim 4\pi k_j^2$ enters as $|S_j|^{1/2} \sim k_j$. This precise relationship requires $|S_j| \sim k_j^2$, which holds for unit-width shells but not for dyadic shells (where $|S_j| \sim 2^{3j} \sim k_j^3$).

Unit-width shells are not a quirky choice. They are the *only* choice that preserves the geometric information the proof exploits. The standard Littlewood–Paley machinery (which uses dyadic shells) is not invoked anywhere in the ACL argument — precisely because Littlewood–Paley discards the lattice geometry that the ACL needs.

4 Convergence: The p-Series vs Geometric Series Distinction

The tail sum $\sum_{j\geq J} k_j^{2-2s}$ controls the Case 2 contribution in the Gronwall closure. Its convergence properties depend on the shell convention, but its convergence *itself* does not.

4.1 Unit-Width Bare Sum

With $k_j = j$:

$$\sum_{j\geq 1} j^{2-2s}$$

is a p-series converging for $2s-2 > 1$, i.e., $s > 3/2$. The integral comparison gives a tail constant

$$C(s) = \frac{1}{2s-3}.$$

At $s = 4$: $C(4) = 1/(2 \cdot 4 - 3) = 1/5 = 0.20$.

The exact full series at $s = 4$ is $\sum_{j=1}^{\infty} j^{-6} = \zeta(6) = \pi^6/945 \approx 1.0173$.

4.2 Dyadic Bare Sum

With $k_j = 2^j$:

$$\sum_{j\geq 0} 2^{j(2-2s)}$$

is a geometric series with ratio $r = 2^{2-2s}$, converging for $2-2s < 0$, i.e., $s > 1$. The closed-form constant is

$$C(s) = \frac{1}{1-2^{2-2s}}.$$

At $s = 4$: $r = 2^{-6} = 1/64$ and $C(4) = 64/63 \approx 1.0159$.

4.3 Why the Thresholds Differ

The thresholds $s > 3/2$ (unit-width) and $s > 1$ (dyadic) differ by a full half-derivative, but **this is not a contradiction.** It reflects the structural difference between p-series and geometric series:

- Up to frequency N, there are $\sim N$ unit-width shells but only $\sim \log_2 N$ dyadic shells.
- The dyadic partition is exponentially sparser, so its bare sum converges more easily.
- Grouping the unit-width sum into dyadic blocks: $\sum_{k=2^{j-1}}^{2^j-1} k^{2-2s} \approx \int_{2^{j-1}}^{2^j} x^{2-2s}\,dx \sim 2^{j(3-2s)}$. The resulting block sum $\sum_j 2^{j(3-2s)}$ converges for $3-2s<0$, i.e., $s>3/2$ — recovering the unit-width threshold exactly.
- The bare dyadic sum with $k_j^{2-2s} = 2^{j(2-2s)}$ omits this internal aggregation factor of $\sim 2^j$ terms per block, which is why its threshold appears lower.

4.4 Numerical Comparison at $s = 4$

Quantity	**Unit-Width**	**Dyadic**
Convergence threshold	$s > 3/2$	$s > 1$
Tail constant $C(s)$ formula	$(2s-3)^{-1}$	$(1-2^{2-2s})^{-1}$
$C(4)$	**0.20**	1.07
Full series ($s=4$)	$\zeta(6) \approx 1.02$	$64/63 \approx 1.02$
Convergence rate	Polynomial (j^{-6})	Exponential (2^{-6j})

Key observation: The unit-width tail constant (0.20) is *five times smaller* than the dyadic tail constant (1.07). The unit-width convention produces a tighter bound. Every downstream constant that absorbs $C(s)$ — including μ in the Gronwall closure — is smaller under unit-width shells. The correction from dyadic to unit-width conventions *improves* the argument.

5 The Invariance Theorem

The bare-sum thresholds differ because they count different objects. When mode multiplicity is properly tracked, both conventions produce the same mathematical conclusion.

5.1 Mode-Weighted Convergence

For unit-width shells, each shell at wavenumber j contributes $\sim j^2$ modes (surface-area scaling on $\mathbb{Z}^3$), each weighted by j^{2-2s}, giving a per-shell total of j^{4-2s}. The p-series $\sum j^{4-2s}$ converges for $2s-4>1$, yielding $s>5/2$.

For dyadic shells, each shell contains $\sim (7\pi/6)\cdot 2^{3j}$ modes (volume scaling), each weighted by $2^{j(2-2s)}$, giving a per-shell total of $2^{j(5-2s)}$. The geometric series $\sum 2^{j(5-2s)}$ converges for $5-2s<0$, again yielding $s>5/2$.

The arithmetic is consistent: the unit-width multiplicity adds 2 to the bare exponent (raising the threshold from 3/2 to 5/2), while the dyadic multiplicity adds 3 to the geometric exponent (raising the threshold from 1 to 5/2, since $(5-2s<0) \Leftrightarrow (s>5/2)$).

5.2 The Intrinsic Threshold

The threshold $s > 5/2$ is an intrinsic property of the Sobolev embedding $H^s(\mathbb{T}^3) \hookrightarrow C^1(\mathbb{T}^3)$ in three dimensions. It is determined by the requirement $s > n/2 + 1$ with $n = 3$, which is a statement about function spaces, not about frequency decompositions.

Invariance Theorem. The shell decomposition convention cannot affect whether a Navier–Stokes energy estimate closes, what the subcritical exponent is, or any other mathematical conclusion. Both conventions yield the same threshold $s > 5/2$ when mode multiplicity is properly tracked. The subcritical exponent $2-s/2$ in the Gronwall closure is determined by the Sobolev substitution $G \leq C_{\text{Sob}} Y^{1/2}$ applied to G^{2-s}, producing $Y^{(2-s)/2} \cdot Y = Y^{2-s/2}$. This is an algebraic identity independent of shell conventions.

5.3 References

The standard subcritical well-posedness result — local existence and uniqueness in $H^s(\mathbb{T}^3)$ for $s > 5/2$ — appears in:

- Majda–Bertozzi, *Vorticity and Incompressible Flow* (Cambridge, 2002), Chapter 3.
- Robinson–Rodrigo–Sadowski, *The Three-Dimensional Navier–Stokes Equations* (Cambridge, 2016).
- Temam, *Navier–Stokes Equations* (AMS Chelsea, 2001).

The Fujita–Kato critical space $\dot{H}^{1/2}(\mathbb{R}^3)$ is determined by scaling symmetry, which is also intrinsic. No frequency decomposition enters these results as anything other than a bookkeeping device.

6 The Corrections Applied

The second form of the shell objection — appendix computations using dyadic constants in a unit-width paper — was a genuine notation inconsistency. The following corrections were applied during the audit:

Location	**Original**	**Corrected**	**Effect**
Appendix C.7	$C(s) = (1 - 2^{2-2s})^{-1}$ (geometric series)	$C(s) = (2s-3)^{-1}$ (p-series)	$C(4)$: $1.07 \to 0.20$
Appendix C.7	$\mu \approx 636\,\nu^3$	$\mu \approx 119\,\nu^3$	Tighter bound
Appendix D.4.10	Geometric series constant	p-series constant	Notation fix
Appendix G.1	$k_j = 2^j$ (dyadic)	$k_j = j$ (unit-width)	Definition fix
Appendix G.7	Def. .25: dyadic shells	Unit-width shells	Definition fix
Appendix G.7	Remark .29: "dyadic"	"unit-width"	Wording fix

Impact on the proof: Zero theorems in the logical chain (Appendix B, rows B1–B21) were affected by any correction. The sums converge under both conventions for $s > 3$. The specific constants differ, but no theorem compares any constant against a numerical threshold. The corrections improve the numerical bounds (μ decreases by a factor of ~ 5) without changing any qualitative or structural conclusion.

7 The Coq `in_shell` Definition

7.1 What the Definition Says

The Coq file `NavierStokesACL.v` contains:

```
Definition in_shell (j : nat) (p : Z3) : bool :=
  (2 ^ j.-1 <= `|norm2Z p|)%N
  && (`|norm2Z p| < 2 ^ j)%N.
```

This tests whether $|p|^2$ falls in $[2^{j-1}, 2^j)$ — a dyadic partition of $|k|^2$. The manuscript's theorem uses unit-width shells $S_j = \{k : j \leq |k| < j + 1\}$.

7.2 Why It Is Non-Load-Bearing

The final theorem `angular_cancellation_lemma` is proved using only the abstract variables `kj`, `Ej`, `shell_list`, `res_list` and the Hypotheses `shell_correct`, `shell_size_bound`, `coll_count`, `trans_count`, etc. The concrete `in_shell` Definition does not appear in the proof term.

When the Section closes, Coq's kernel generalises the theorem over only the declarations it actually depends on. The official documentation states: "all global objects defined in the section are generalized with respect to the variables and local definitions *they each depended on* in the section" (Coq 8.18 Reference Manual, Sections). The phrase "they each depended on" is the critical qualifier — generalisation is per-object, not per-section.

7.3 Mechanical Verification

- `Print Assumptions angular_cancellation_lemma` does not mention `in_shell`.
- `Check angular_cancellation_lemma` displays the full generalised type, showing that `in_shell` is absent.
- The theorem is universally quantified: $\forall R$, N_{sh}, k_j, $E_j, \ldots$ It is a theorem about *any* shell structure satisfying the stated hypotheses.

This is guaranteed by the Curry–Howard correspondence: every Coq theorem is a term in the Calculus of Inductive Constructions, and that term explicitly encodes every definition, variable, and hypothesis it depends on. If `in_shell` does not appear in the proof term, the theorem is as independent of `in_shell` as it is of any other definition in the Coq environment.

The French cybersecurity agency ANSSI mandates exactly this check: "the `Print Assumptions` command shall be used in order to determine which constants lacking a definition body have been used to prove key theorems" (ANSSI, *Requirements on the Use of Coq in the Context of Common Criteria Evaluations*).

7.4 Precedent in Major Formalisations

The coexistence of concrete definitions with parametric theorems is the dominant architectural pattern in large-scale formal verification:

- **CompCert** (∼100,000 lines of Coq): utility functions in `Coqlib` coexist with parametric compilation correctness theorems.
- **Feit–Thompson** (∼170,000 lines of Coq): concrete group instances coexist with abstract theorems about arbitrary finite groups.
- **Mathlib** (∼2 million lines of Lean): the entire library is built on parametric typeclasses with concrete instances registered separately.
- **Flyspeck**: contains an explicitly labelled directory called `informal_code` described as "not required for the flyspeck project."

Definitive statement. The `in_shell` definition could define shells as cubes, as random subsets, or as the empty set. It would not change a single `Qed` in the file, because no proved theorem computes from `in_shell` directly. There is no mismatch. There is a definition that is not load-bearing.

8 The AI Auditing Failure

8.1 Empirical Record

The shell convention was flagged as a concern in every audit configuration tested:

Session	What Was Flagged	System(s)	Rounds to Resolve
Initial ACL audit	`in_shell` dyadic vs. unit-width	All three	1–4
Section 6 review	Case 2 tail sum uses dyadic constant	System B	1
Remark 9.7 review	Subcritical exponent "wrong" due to shell convention	System B	1
Appendix C review	C.7 geometric series constant	Systems A, B	1
Appendix G review	G.1 shell definition	System B	1
Coq file review	`in_shell` mismatch (again)	All three	1–3

Total instances: ≥ 10 across all sessions. No other single objection appeared more than twice.

8.2 The Failure Modes Operating

Three documented failure modes from the Common Failure Modes taxonomy (Parts 1–4) were operating:

FM 12: Convention Mismatch Escalation

The system identifies a notation difference between two representations of the same mathematics and escalates it to a mathematical error. The $Z_{\text{Coq}} = \nu \cdot Z_{\text{manuscript}}$ convention and the `in_shell` dyadic-vs-unit observation both triggered this mode. In each case, the system correctly identified a notational difference and incorrectly concluded that a theorem was affected.

FM 16: Pattern Overfit (Prior Suspicion Reuse)

Once a system identified one legitimate shell convention issue (e.g. the C.7 geometric series constant), it began pattern-matching the same suspicion onto every subsequent section — including Remark 9.7, where the subcritical exponent $2 - s/2$ arises from the Sobolev substitution and has nothing to do with shell conventions.

Proximity Bias (Documented in AI Literature)

The GSM-Symbolic benchmark (Apple Research, ICLR 2025) demonstrated that adding a single clause that *seems* relevant to a question but does not contribute to the reasoning chain causes performance drops of up to 65% across all state-of-the-art models. An AI system that encounters `in_shell` (mentioning shells, using frequency-band notation) in the same Section as a theorem about shells infers a logical relationship from textual co-occurrence. This is syntactic pattern-matching, not structural dependency analysis.

The miniCTX benchmark (ICLR 2025) showed that LLM-based provers are typically "unaware of new theorems and definitions" — they lack the capacity to trace dependency graphs through a type-theoretic kernel. The PALM study (Purdue, 2024), which analysed 520 proof-generation errors, found the dominant failure mode was "correct high-level structure with wrong low-level details" — models can identify that a proof involves shells but cannot determine which specific definitions are load-bearing.

8.3 The Compounding Effect

The shell convention objection is uniquely persistent because it combines three failure triggers:

1. A *visible* notational difference (the word "dyadic" appears in the Coq file, the word "unit-width" appears in the paper).
2. A *plausible* mathematical concern (shell conventions do affect bare-sum convergence thresholds, even though they don't affect mode-weighted thresholds).
3. A *concrete* code artifact (the `in_shell` definition has specific line numbers, specific syntax, and specific numerical constants that can be quoted).

This combination makes the objection feel substantive even when it is not. A system that says "line 134 defines dyadic shells but the theorem uses unit-width" sounds like it has found

something specific. The specificity is real — the *relevance* is not. But distinguishing specificity from relevance requires understanding the parametric architecture, which current AI systems cannot do from the code alone.

9 Definitive Resolution

The shell convention question is closed. The following five statements are established by the evidence in this document and the references cited:

1. The manuscript uses unit-width shells because the proof requires lattice-resolution geometry. Dyadic shells would lose the geometric information the proof exploits. This is a mathematically necessary choice, not a convention preference.
2. The appendix computations that used dyadic constants have been corrected to match. Six locations were updated (C.7, D.4.10, G.1, G.7, Definition .25, Remark .29). The corrections *improve* the numerical bounds without affecting any theorem.
3. The Coq `in_shell` definition is non-load-bearing. This is guaranteed by the Curry–Howard correspondence, verified by `Print Assumptions`, and standard across every major formalisation project.
4. The subcritical threshold $s > 5/2$ is convention-independent. Both unit-width and dyadic decompositions produce the same threshold when mode multiplicity is properly tracked. The subcritical exponent $2 - s/2$ is determined by the Sobolev embedding, not by shell conventions.
5. No theorem in the logical chain (Appendix B, rows B1–B21) is affected by the shell convention choice. The proof closes identically under both conventions.

Appendix: Source Summary

Lattice Counting

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